

KHUSH TRIVEDI

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SUMMARY

AI & ML Engineering student (CPI 9.0) with hands-on experience building agentic AI systems, RAG pipelines, and LLM applications using LangChain, LangGraph, FastAPI, ChromaDB, and FAISS. Experienced across the full stack: backend APIs, database design, and frontend development (PostgreSQL, Redis, Docker, React).

TECHNICAL SKILLS

- **Programming:** Python, SQL, C
- **AI & Generative AI:** LLMs, Generative AI, AI Agents, RAG, Prompt Engineering
- **LLM Frameworks & APIs:** LangChain, LangGraph, Groq API, Gemini API, OpenRouter, Hugging Face Transformers
- **Retrieval & Vector Databases:** ChromaDB, FAISS, Embeddings
- **Backend & APIs:** FastAPI, REST APIs, CRUD Operations
- **Databases:** PostgreSQL, MySQL, SQLite, Redis
- **Machine Learning & NLP:** Scikit-learn, Pandas, NumPy, NLTK, spaCy, Classification, Regression, Feature Engineering
- **Frontend:** React, Streamlit, Tailwind CSS
- **Developer Tools:** Git, GitHub, Docker, VS Code, Jupyter Notebook

EXPERIENCE

Agentic AI Intern | *AgenticX AI Labs* | *Virtual Internship* | 2026 | [Certificate](#)

- Built multi-step tool-using agents with LangGraph, designed agentic AI workflows, and implemented autonomous AI orchestration patterns.

Machine Learning Intern | *Auspify Technologies* | *Virtual Internship* | 2026 – Present | [Offer Letter](#)

- Developing and evaluating predictive ML models and building end-to-end machine learning pipelines on real-world applied projects.

PROJECTS

AI Research Agent – Agentic Web Research System | *Python* | *LangGraph* | *FastAPI* | *LLM Tool-Calling* [GitHub](#)

- Built a conversational research agent with LangGraph that routes each question to the right tool (Tavily web search or Open-Meteo weather lookup) via LLM tool-calling, maintaining agent state across multi-step reasoning iterations until a final answer is produced.
- Implemented persistent per-user conversation history via LangGraph's SqliteSaver checkpoint, multi-user auth with bcrypt-hashed passwords and JWT session tokens, and a Streamlit chat UI with token-by-token streaming responses and a sidebar for resuming past conversations.

Blog Writing Agent – Multi-Step AI Content Generation System | *Python* | *LangChain* | *LangGraph* | *Prompt Engineering* [GitHub](#)

- Built a multi-step, LLM-driven pipeline that takes a topic request and progresses it through planning, structured outline generation, drafting, and content refinement stages.
- Designed the workflow to pass state/context between stages (outline to draft to refined output) rather than generating content in a single LLM call.

DocTalk– RAG-Based Document Intelligence System | *Python* | *FastAPI* | *LangChain* | *ChromaDB* | *Groq API* | *Streamlit* [GitHub](#)

- Built a full RAG pipeline: extracted text from uploaded PDFs, chunked content, generated embeddings, and indexed them in ChromaDB for semantic retrieval.
- Integrated a FastAPI backend with a Groq-hosted LLM to generate context-grounded answers from retrieved chunks, with fallback handling for out-of-document queries.

YouTube RAG Chatbot – Video Knowledge Assistant | *Python* | *LangChain* | *FAISS* | *OpenRouter* | *Streamlit* [GitHub](#)

- Built a RAG-based conversational assistant that ingests YouTube video transcripts, chunks the transcript text, and generates embeddings using BAAI/bge-m3.
- Indexed and retrieved transcript chunks via FAISS, passing relevant context to an LLM (via OpenRouter) to produce context-grounded, conversational answers.

URL Shortener – Full-Stack System Design Project | *Python* | *FastAPI* | *PostgreSQL* | *Redis* | *Docker* | *Nginx* | *React* [GitHub](#)

- Architected a 4-tier containerized system (Nginx reverse proxy, FastAPI, PostgreSQL, Redis) via Docker Compose, implementing Base62 encoding on auto-incremented IDs and indexed database lookups.
- Implemented a Redis cache-aside strategy for high-throughput redirects and built REST APIs for link creation, redirection, and click analytics, with support for custom aliases and link expiration.

EDUCATION

New LJ Institute of Engineering and Technology, GTU – Ahmedabad

2023 – Present

B.E. – Computer Science Engineering (AI & ML) | CPI: 9.0

ACHIEVEMENTS

- TechFest, IIT Bombay – College Ambassador; Certificate of Appreciation (Rank under 20,000) [link](#)
- HackOvate Hackathon – Participant [link](#)
- Mind Sprint Hackathon (Unstop) – Participant [link](#)

CERTIFICATIONS

- NPTEL – Natural Language Processing [link](#)
- Introduction to Data Science – University of Michigan (Coursera) [link](#)
- Advanced Object-Oriented Programming using Python – New LJ Institute of Engineering and Technology [link](#)